

REDMOND, OR, USA

WMO: 726920

Lat: 44.256N

Lon: 121.139W

Elev: 3043

StdP: 13.15

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24230

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.1	12.4	-0.3	6.0	9.7	5.7	8.2	18.3	24.7	44.5	21.7	43.5	4.2	330	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.8	93.6	61.3	90.7	60.6	87.8	59.7	63.8	88.4	62.2	86.0	60.7	83.5	10.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.9	71.8	67.0	53.0	67.0	66.8	51.2	62.6	66.6	30.4	87.9	29.2	85.5	28.1	83.6	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.7	16.8	-3.9	99.5	8.2	3.1	-9.8	101.8	-14.6	103.6	-19.2	105.4	-25.2	107.7
			-4.5	66.5	8.1	1.7	-10.3	67.7	-15.1	68.6	-19.6	69.6	-25.4	70.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	34.0	35.5	40.3	44.5	52.3	58.9	67.4	66.1	58.2	47.3	38.7	32.2		
	DBStd	14.09	8.70	7.98	6.61	6.70	7.43	7.08	6.25	5.98	7.10	7.26	8.93	9.06		
	HDD50	2524	496	408	306	192	60	9	0	1	12	136	349	555		
	HDD65	6456	960	826	764	616	398	206	48	59	225	548	788	1018		
	CDD50	1803	1	1	6	26	132	277	538	500	257	54	10	1		
	CDD65	264	0	0	0	0	5	25	121	94	19	0	0	0		
	CDH74	7218	0	0	1	39	275	839	2855	2297	855	57	0	0		
	CDH80	3135	0	0	0	4	74	301	1390	1047	311	8	0	0		
(14) Wind	WSAvg	6.4	5.7	6.2	7.0	7.0	7.0	7.1	6.6	6.2	5.9	5.7	6.1	5.9		
(15) Precipitation	PrecAvg	8.6	1.1	0.6	0.7	0.6	0.8	0.7	0.4	0.5	0.4	0.6	1.1	1.1		
	PrecMax	12.4	3.1	2.4	1.9	1.7	2.3	3.9	3.7	2.5	1.3	1.9	3.1	5.5		
	PrecMin	5.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	1.7	0.6	0.5	0.5	0.5	0.6	0.8	0.7	0.6	0.4	0.4	0.8	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	63.5	70.3	79.2	87.5	93.4	98.7	97.7	92.2	80.1	68.9	57.5		
		MCWB	46.0	46.7	50.2	54.9	58.8	62.7	64.0	61.8	60.1	55.0	51.3	45.6		
	2%	DB	54.2	56.6	64.1	72.5	81.3	87.6	94.6	93.1	87.9	74.3	62.5	52.5		
		MCWB	44.6	45.0	47.1	52.3	56.8	60.4	61.9	60.9	58.7	52.6	49.4	44.1		
	5%	DB	50.0	52.3	59.2	66.5	75.9	83.0	91.3	89.9	83.3	69.3	57.1	48.8		
		MCWB	42.0	42.7	45.0	49.3	54.4	58.3	60.9	60.3	57.5	51.1	46.4	42.2		
	10%	DB	46.4	48.3	54.7	61.1	71.5	78.8	88.3	86.3	79.0	63.8	53.0	45.6		
		MCWB	40.0	40.3	43.9	46.8	52.8	55.9	60.0	59.2	55.7	49.4	44.3	39.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	49.3	52.2	56.5	61.0	64.4	66.6	65.4	62.9	57.4	54.1	48.5		
		MCDB	56.3	61.5	64.2	75.1	81.9	88.2	93.6	90.0	85.9	73.2	64.3	54.8		
	2%	WB	45.0	46.1	49.5	53.2	58.2	61.4	64.1	63.3	60.3	54.5	50.1	45.2		
		MCDB	52.6	55.5	57.9	70.9	76.7	84.0	89.5	87.6	82.2	69.8	60.3	51.0		
	5%	WB	42.4	43.4	47.1	50.4	56.1	59.4	62.4	61.7	58.6	52.3	47.5	42.5		
		MCDB	49.0	51.3	58.5	64.9	73.1	80.1	87.0	85.1	79.8	66.3	56.6	48.3		
	10%	WB	40.4	40.9	44.4	47.8	53.8	57.3	60.9	60.1	56.7	50.2	44.5	40.1		
		MCDB	46.2	47.4	54.9	59.8	68.7	76.0	84.6	82.6	76.9	62.7	51.8	45.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	20.8	25.3	27.8	29.6	32.6	37.8	37.3	35.8	29.6	21.7	18.1		
		MCDBR	23.1	27.0	34.0	39.6	39.0	40.0	41.8	41.9	43.5	39.3	28.1	21.3		
		MCWBR	14.8	16.9	19.9	21.4	18.3	17.2	16.5	17.2	19.9	21.1	16.9	14.3		
	5% WB	MCDBR	20.8	24.5	31.3	36.0	34.8	37.4	38.6	38.7	39.5	33.3	25.9	19.0		
		MCWBR	14.1	15.8	20.2	19.9	16.6	16.3	15.8	16.2	18.6	18.7	16.0	13.3		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.288	0.289	0.305	0.323	0.317	0.331	0.321	0.304	0.301	0.279	0.280		
	taud		2.542	2.559	2.578	2.529	2.495	2.539	2.507	2.551	2.571	2.570	2.593	2.548		
	Ebn at Noon		272	291	304	305	300	301	296	296	295	282	272	262		
	Edh at Noon		21	24	27	31	33	32	33	30	27	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		495	780	1159	1626	1986	2305	2473	2120	1616	1006	572	403		
	RadStd		58	94	98	138	149	125	101	114	89	82	56	56		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+67

Nomenclature: See separate page